

Executive Project Blueprint: Multi-School SaaS ERP System

1. Project Overview & Vision

This project is an enterprise-grade, Multi-Tenant SaaS ERP Platform designed to digitize and automate complete school operations. It replaces manual, paper-based management with a fully transparent, scalable, and workflow-driven digital ecosystem. While the system hosts multiple schools centrally, each school operates within a strictly isolated, secure environment.

2. Core Architecture & Design Principles

- **Multi-Tenant SaaS Isolation:** A single centralized database infrastructure where every school's data is partitioned and completely invisible to others.
- **Role-Based Access Control (RBAC):** Security is foundational. Users only see and interact with data necessary for their specific job functions.
- **Workflow-Driven Operations:** Modules are not isolated data-entry forms; they are connected pipelines. (e.g., An admission approval automatically triggers a fee invoice).
- **Audit-First Design:** Every critical action in the system is logged with user details, timestamps, and IP addresses to prevent fraud and ensure total accountability.

3. The Role System & Access Model

The platform mirrors a real-world school hierarchy. Access and capabilities are strictly governed by these roles:

- **Super Admin (Platform Owner):** Controls the SaaS infrastructure, manages school subscriptions, configures global settings, and oversees system-wide backups.
- **School Admin (Principal):** The "Command Center" for a single school. They hold full read-access for their institution, approve admissions, manage staff assignments, and oversee fee structures.

- **Academic Coordinator:** Handles structural academic setup, including curriculum mapping, subject assignment, and timetable generation.
- **Admission Staff:** Manages the intake pipeline, processes applications, verifies applicant documents, and shortlists candidates for approval.
- **Fee Officer:** Controls the financial operations, generates invoices, tracks payments, applies authorized scholarships, and handles receipts.
- **Teacher:** The academic execution layer. Responsible for daily student attendance, entering examination marks, and student evaluations.
- **Parent:** The transparency layer. Gains secure access to track their child's attendance, view report cards, and monitor fee balances.
- **Auditor:** The security monitor. Holds read-only access to system-wide audit logs, financial modifications, and login histories to ensure compliance.

4. Core Modules & Functionalities

- **Admission Management:** An automated pipeline moving from Online Application → Document Verification → Shortlisting → Scholarship Linking → Final Admin Approval → Automated Student Creation.
- **Student & Academic Management:** Acts as the single source of truth for student data. Manages class/section assignments, daily attendance, examination grading, report card generation, and end-of-year promotions.
- **Fee & Scholarship Management:** Controls the financial health of the school. Features automated invoice generation, payment tracking, mobile money integration (MTN/Airtel), and a strict workflow for applying scholarships and fee discounts.
- **Staff & HR Management:** Tracks all human resources, managing staff profiles, daily staff attendance, department allocations, and teaching workloads.
- **Operational Modules:** Includes Library Management (book issues/returns and fines) and Sports/Activities Management (event participation and team rosters).
- **Communication & Dashboards:** Role-specific dashboards (Admin, Teacher, Parent) combined with automated SMS and Email notification engines for alerts and updates.

5. Data Governance & CRUD Strategy (Critical Feature)

Standard applications allow users to freely Create, Read, Update, and Delete (CRUD) data. Our ERP strictly forbids this to ensure absolute data integrity. **CRUD is never direct; it is workflow-controlled.**

- **Create (Workflow Validation):** Records are generated through processes, not manual entry. For instance, a student is officially "Created" in the database only after the School Admin grants final admission approval.
- **Read (Siloed Views):** Users only access their jurisdiction. A teacher sees only their assigned class; the School Admin sees the entire school; the Super Admin sees platform-wide metrics.
- **Update (Audited Edits):** Modifying locked data (like published exam results or paid fee structures) requires a formal request and approval workflow. Every single update generates a permanent audit trail.
- **Delete (Zero Hard Deletes):** This is a strict database rule. Users, including the School Admin, cannot permanently delete records. Instead, the system uses **Soft Deletes**. When a student transfers or a teacher resigns, their status is simply updated to Inactive, Archived, or Transferred. This preserves all historical financial and academic data for future audits.

6. School Admin CRUD Capabilities

While the School Admin has the highest authority within their institution, their powers are governed by the system's safety rules:

- **Admin Create:** They hold direct creation power for registering new staff members and teachers.
- **Admin Read:** They possess a 100% unrestricted view of their school's data (fees, academics, staff logs).
- **Admin Update:** They can override fee structures, reassign staff roles, and update student statuses, but every action is permanently logged against their user ID.
- **Admin Delete:** They can only "Deactivate" users, utilizing the Soft Delete architecture to maintain database stability.

7. Deployment Strategy & Business Value

- **Phased Development:** We will deploy the Database & Security Architecture first, followed by the Core Modules (Admissions, Academics, Finance), and finally the Communication and Reporting layers.
- **Scalability:** The REST API architecture ensures the platform can easily scale from hosting one school to thousands without performance degradation.
- **Client Value Proposition:** We are delivering a highly secure, audit-proof, and fully automated system that directly mirrors physical school workflows while completely eliminating paper-based inefficiencies and financial blind spots.